Electromagnetic Field Microphone

By soundwise in CircuitsElectronics

Introduction: Electromagnetic Field Microphone

An electromagnetic microphone is an unconventional tool for sound designers, composers, hobbyists (or ghost hunters). It is a simple device that uses an induction coil to capture and convert Electro-Magnetic Fields (EMF) into audible sound. There are some commercial ones available, such as the Elektrosluch which is strong enough to capture ambient EMFs and some cheaper ones that need to be close enough to the source of the EMF.

This tutorial demonstrates how to make such a microphone, that captures the environmental EMF and converts it into audible sound by using an amplifier. It may not have the quality of an Elektrolusch but is still a fun project with which you can get some interesting sound effects (See videos in Step5).

The project is suitable for beginners. I made it having very little experience with circuitry and soldering myself.

Materials:

- An old pair of headphones.
- Enameled copper wire.
- Headphone amplifier. I used the TDA1308 but you can try also with other models.
- Battery holder. This depends on the amplifier you choose. If it is the model mentioned above, you are going to need a 6V holder. I recommend choosing one with an on/off button to make your life easier. Example here
- Sandpaper or nail file.
- 4 x AA batteries

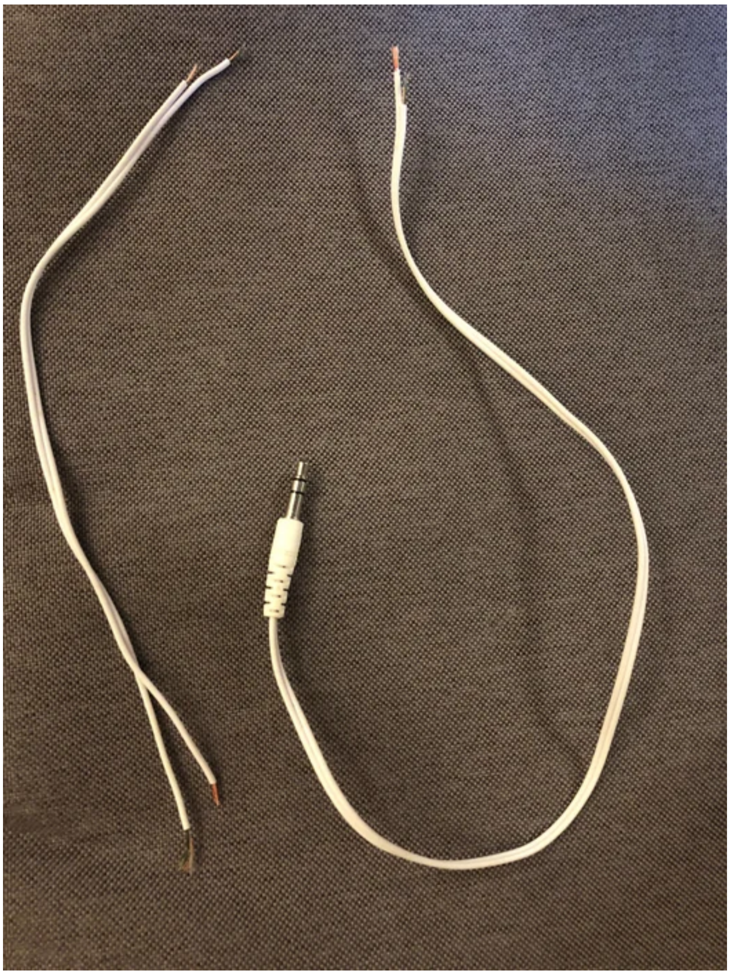
Tools:

- Soldering Iron and solder wire.

Optional:

- Access to a 3D-printer. This is to create a protective case for the copper wire. You can download a design here, but you need to edit the dimensions depending on the size of the copper wire holder.
- Tube shrinkers to cover as much as possible of the naked wires. This will reduce unwanted noise significantly.
- A 125*80*32mm case to put the amplifier and batteries in.
- Glue gun.

Step 1: Prepare the Connections



Since the wire is enamelled, you will need to file the two ends using the sandpaper/nail file.

Next, you need to cut your old headphones.

You are going to need two pieces of cable: one that includes the phone jack on its one end and one bare cable.

You can use the sandpaper/nail file to get rid of the covering.

Step 2: Protecting Your Microphone (optional)

If you have purchased the optional case for the batteries and amplifiers, you need to drill two holes and squeeze the cables through them before you keep on with soldering.

If you haven't got a driller, like in my case, you can also try with a needle and a hammer. The plastic in these cases is usually thin and it should not take long to drill through.

Once you have drilled two holes, pass through the two cables from the previous steps.

Use the photo of the ready product as a guide.

Step 3: Solder the Connections

Follow the pictured schematic to solder the connections (obviously, designing is not one of my strengths...)

The red wire on headphones is usually for Right, the copper for Ground and a black, white or green for Left. If the headphones you used had a microphone, you will find an additional wire (just ignore this wire, no need to connect anywhere).

Don't worry which end to use between the copper wire for Left and Right, as it doesn't matter.

For the ground, you can either solder it along the Left wire or leave it unconnected.

If you don't want to follow the optional steps for housing, just add the 4 x AA batteries and you are ready to go!
You can connect the jack to any recorder or computer.

Step 4: Protecting Your Microphone (Optional)

If you have followed step 2, you should now have a similar result as in the picture.

Try to squeeze in the battery housing and the amplifier. You can use some glue to keep the batteries separated from the amplifier as in the picture.

For the copper wire, either use the 3D printed case or find a cap from a bottle that fits the enamelled copper wire.

You can use a glue gun to ensure the case won't come off, but also to make the microphone waterproof.

Close the case, connect the jack on a recorder and you are ready to go!

Step 5: Have Fun!

Once you connect your microphone to a recorder, turn on the battery power.

You should be able to hear your surrounding electromagnetic noise!

Note: If you are using headphones, please make sure to have turned down the volume of your recorder before wearing them as the noise from the microphone can become too loud and damage your hearing.

Ideas:

Try to use your microphone close to electronic sources, such as washing machine, lights, computer, phone, wifi router.

Did you know that you can hear noises with an EMF microphone underwater? Check out the videos!

Digital Processing:

The captivating thing about recording EMF is that the frequency spectrum you will get is by far richer than the one you'll get from a normal microphone (check the attached spectrogram).

This means that you can use extreme pitch shifts and get results that sound completely different from your original recordings.

Check out an example of some extreme pitch-shifted emf recordings on the attached .wav files, with which you can make music!